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Water quality prediction using enhanced parrot optimization algorithm with ensemble learning techniques

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Abstract

Water quality monitoring has become increasingly critical as urbanization and industrialization escalate global water pollution challenges, contributing to approximately 1.8 million deaths annually from waterborne diseases according to World Health Organization estimates. This study introduces a novel integration of the Parrot Optimization Algorithm (POA) with ensemble learning (ANN, RF, XGBoost) for water quality prediction using the Cauvery River dataset (from 2018 to 2022, (642 samples, 26 parameters)). The approach was validated using a comprehensive dataset from the Cauvery River collected by the Tamil Nadu Pollution Control Board. Experimental results demonstrate that POA significantly outperformed traditional optimization techniques, achieving 85.5% accuracy, 86.3% precision, 86.2% recall, and 87.3% F1 score. Crucially, we have statistically validated that POA exhibited 32% faster convergence and selected 50% more relevant features compared to Particle Swarm Optimization (PSO) and Genetic Algorithm (GA). This research contributes a computationally efficient and highly accurate decision-support tool that enables stakeholders and policymakers to monitor water pollution in real-time, facilitating early intervention strategies and promoting sustainable water resource management practices globally.

Keywords Water quality, Ensemble, Feature selection, Parrot optimization, Prediction

Introduction

River water quality monitoring represents a critical global health imperative, directly impacting human well-being through its connection to waterborne disease transmission and ecosystem sustainability. The World Health Organization estimates that 1.8 million people died from waterborne diseases in 2018 alone [1]. This alarming figure underscores the necessity of continuous surveillance and evaluation of water bodies against established maximum permissible concentration (MPC) thresholds for

various contaminants. When pollutant concentrations exceed these critical limits, immediate remedial actions must be implemented to restore water quality and mitigate contamination levels. Such monitoring protocols are essential for determining water suitability for diverse applications including potable water supply, fishing activities, and recreational purposes, while ensuring compliance with regulatory frameworks [2].

Environmental researchers have increasingly focused on river water quality assessment as a fundamental indicator for forecasting potential disease outbreaks. Over recent decades, the scientific community has employed numerous computational approaches to analyze water quality parameters and generate reliable disease predictions. However, the effectiveness of these predictive models is predominantly contingent upon the quality and characteristics of the training datasets [3]. These datasets typically encompass physical, chemical, and biological

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measurements that collectively represent the overall condition of the water ecosystem. In numerous instances, suboptimal feature selection can significantly diminish model accuracy and compromise predictive capabilities [4]. Consequently, feature selection methodologies have become instrumental in identifying the most influential variables and enhancing overall model performance [5].

Feature selection remains a critical issue in water quality prediction, as high-dimensional datasets often contain redundant or irrelevant parameters that can degrade model performance. Traditional optimization algorithms such as PSO and GA have shown limitations in handling the complex search spaces associated with water quality feature selection, often suffering from parameter sensitivity and slow convergence [6]. The Parrot Optimization Algorithm (POA) was selected for this study due to its unique characteristics that make it particularly suitable for feature selection in water quality assessment. Unlike traditional metaheuristic algorithms, POA incorporates several bio-inspired mechanisms that provide advantages in terms of exploration–exploitation balance, convergence speed, and solution quality. Table 1 compares POA with other popular heuristic algorithms in the context of feature selection for water quality prediction.

POA's unique combination of Lévy-flight foraging, which enables large jumps in the search space to escape local optima, and communication-intensity scaling, which facilitates efficient information sharing between solutions, makes it particularly well-suited for navigating the complex feature landscape of water quality data. Additionally, POA's fear response mechanism helps maintain population diversity, preventing premature convergence a common issue in traditional algorithms when dealing with high-dimensional datasets.

Based on the above research gaps, the proposed work involves the following major contributions which are highlighted as follows: Firstly, the proposed work utilized a comprehensive Cauvery River Dataset (2018–2022) which is acquired from the TNPCB (Tamil Nadu Pollution Control Board) facilitating the real-world data. Secondly, it utilizes KNN-based machine learning technique thereby preserving the spatial/temporal

correlations with simple mean or median approach to clean the missing value and to improve the prediction accuracy. Thirdly, the proposed work involves the use of the Parrot Optimization Algorithm (POA) for feature selection, leveraging a trade-off between the exploration and exploitation phases. Additionally, it achieves speedy convergence when compared with the traditional metaheuristic algorithms such as particle swarm optimization (PSO), genetic algorithm (GA) and grey-wolf optimization (GWO). Finally, a hybrid ensemble architecture has been developed that integrates artificial neural network (ANN), random forest (RF) and XGBoost (Extreme Gradient Boosting) approaches through stacking and voting, leading to an accurate and computationally efficient decision-support tool for river water quality assessment.

Organization of the paper

The overall structure of this study is organized as follows: Section "Fitness Evaluation: The fitness of each parrot is evaluated using a fitness function that considers both classification accuracy and feature subset size" provides a review of related works. Section "Position Update: The positions of parrots are updated using the Eqs. (3, 4, 5 and 6) based on their foraging, communication, and fear behaviors." introduces the inspiration behind the POA and Ensemble technique model. Section 4 presents the experiment results and, Section "Performance Analysis" concludes the study and suggests ideas for future research.

Related works

The domain of water quality prediction has witnessed a remarkable evolution over the past decade, progressing beyond conventional statistical methods to embrace a spectrum of sophisticated computational methodologies. Initial efforts were predominantly rooted in traditional statistical analyses and basic regression models designed to correlate water quality parameters with pollution indicators. However, these foundational techniques were fundamentally limited in their ability to model the intricate, non-linear dynamics that characterize aquatic ecosystems. The emergence of machine learning has catalysed

Table 1 Comparison of heuristic algorithms for feature selection in water quality prediction

Algorithm	Exploration mechanism	Exploitation mechanism	Convergence speed	Parameter sensitivity	Suitability for high-dimensional data
POA	Lévy-flight foraging and fear response	Communication-intensity scaling and stationary behavior	Fast	Low	High
PSO	Velocity-based movement	Personal and global best positions	Moderate	High	Moderate
GA	Crossover and mutation	Selection of fittest individuals	Slow	High	Low
GWO	Pack hunting behavior	Alpha, beta, and delta wolves lead the pack	Moderate	Moderate	Moderate

a paradigm shift, empowering researchers to generate predictions with unprecedented accuracy and depth. The following review offers a chronological syntheses of these methodological advancements, culminating in an articulation of the persistent research voids that this study aims to address.

A foundational comparative investigation by Barzegar et al. [7] engineered a hybrid model that merged wavelet transforms with extreme learning machines to achieve an accuracy of 80.9%. This approach demonstrated the value of preprocessing techniques to improve model performance by extracting meaningful features from complex data. The disadvantage of their method is that the selection of appropriate wavelet parameters can be complex and may require domain expertise. Offering a different perspective, Avila et al. [8] used five machine learning (ML) models to predict *E. coli* levels in the Oreti River, located in Wallacetown, New Zealand. The probabilistic framework that furnished critical insights into prediction uncertainty an aspect frequently neglected in deterministic models and attained an accuracy of 83.7%. The model's ability to quantify uncertainty makes it valuable for risk assessment in water management. The major disadvantage of their approach is that the computational complexity can increase significantly with a larger number of variables, making it less scalable for high-dimensional datasets.

The period spanning 2020 to 2022 was distinguished by a diversification of methodologies, as the research community explored a broader array of models from advanced neural networks to complex ensembles often in pursuit of near-perfect predictive accuracy. Nayak et al. [9] engineered ANN, effectively capturing time-series dependencies to achieve 84.3% accuracy. The suggested method's disadvantage is that LSTM models require large amounts of sequential data to be trained effectively, which may not always be available in environmental monitoring contexts. In parallel, Jiang et al. [10] introduced a deep convolutional feature fusion network that integrated multiple data sources, attaining 82% accuracy through the effective combination of spatial and temporal features. This highlighted the importance of multi-source data integration. The suggested method's disadvantage is its architectural complexity, which makes it difficult to implement and optimize without specialized expertise in deep learning. This era was also defined by the creation of increasingly complex and high-performing ensembles. The pursuit of peak performance culminated in studies such as Wong et al. [11], who reported a perfect accuracy of 95% using a stacked model combining multiple algorithms, and Khullar and Singh [12], who obtained 94% accuracy using a BiLSTM model. The former study represented the pursuit of peak performance through highly complex ensembles. The suggested

method's disadvantage is its extreme complexity, which makes it computationally expensive, difficult to reproduce, and potentially prone to overfitting. A significant shift in recent research has been a movement beyond merely achieving high accuracy to ensuring model robustness, interpretability, and computational efficiency. In 2023, the power of deep learning to capture complex spatiotemporal patterns was demonstrated by Guo et al. [13], who developed a novel hybrid CNN-LSTM model to predict the WQI in Hainan Province, China. Their model exhibited superior performance over standalone models, achieving a high coefficient of determination ($R^2=0.98$). The suggested method's disadvantage is that the hybrid architecture's performance is highly dependent on the quality and quantity of training data, and it may not perform well on data from different geographical regions. A pivotal development in 2023 was the integration of Explainable AI (XAI).

Bui et al. [14] introduced an integrated framework that coupled ensemble models (AB-ADTree model) technique for groundwater potential mapping. This research was instrumental in demystifying the 'black box' nature of complex models, making the results highly actionable for policymakers. The suggested method's disadvantage is that while SHAP provides local explanations, global interpretation of the entire model's behavior can still be challenging. This trajectory towards more sophisticated and interpretable models accelerated into 2024. Aderemi et al. [15] proposed an XAI method specifically for forecasting pollution in drinking water, further cementing the importance of transparency. Their proposed method's disadvantage is that the implementation of XAI techniques can add a layer of computational overhead to the modelling process. Goodarzi et al. [16] applied three models to predict groundwater quality, with Multivariate Adaptive Regression Splines achieving an exceptional 99.61% accuracy. This study showcased the continued high performance of advanced boosting models in water quality prediction tasks. The suggested method's disadvantage is that achieving such high accuracy might indicate overfitting to the specific dataset, and the model's performance on new, unseen data needs careful validation. Saberioon et al. [17] focused on chlorophyll-a, a key eutrophication indicator in Třeboň Basin, finding that random forest provided the most accurate and stable predictions ($R^2=0.89$). This suggested the strong potential of Random forest for lake water quality management. Since the proposed method is based on tree-based models, it is highly sensitive to hyperparameter settings, requiring a systematic tuning process to achieve optimal results. Representing the cutting edge of architectural innovation, more recently, Zhu et al. [18] introduced a hybrid deep learning model for predicting turbidity (UT)

and dissolved oxygen (DO) in river waters. By combining a CEEMDAN-CNN-LSTM, this new hybrid model improves the forecast accuracy at peak locations while reducing the average prediction errors. The computational complexity of the suggested method and the need for large datasets are its disadvantages. Similarly, Chen et al. [19] have demonstrated river water quality prediction with machine learning techniques. The study analyzes and forecasts chlorophyll-a (Chla) using an innovative data fusion framework based on Bayesian model averaging (BMA) by combining several deep learning models. The findings confirmed that the suggested approach (BMA) outperformed other individual methods. One disadvantage of the suggested approach is that parameter adjustment might be expensive and time-consuming.

Synthesis of research gaps and motivation

Critically, a comprehensive synthesis of the literature reveals that despite these methodological strides, persistent weaknesses limit the effectiveness of current approaches and delineate clear research opportunities. The following key gaps have been identified:

- i) Suboptimal feature selection methodologies:
- ii) Feature selection remains a formidable challenge, as high-dimensional datasets frequently contain redundant or irrelevant parameters that can impair model performance. Conventional optimization algorithms, such as Particle Swarm Optimization (PSO) and Genetic Algorithm (GA), have demonstrated inadequacies in navigating the complex search spaces inherent to water quality feature selection, often hampered by parameter sensitivity issues and protracted convergence rates.
- iii) Limited exploration of sophisticated ensemble frameworks: the majority of existing studies have concentrated on individual machine learning models or simplistic ensemble approaches (e.g., voting). The considerable potential of advanced, dynamic ensemble architectures, such as stacking which can dynamically learn optimal weights for model combination—remains largely untapped in the water quality prediction literature.
- iv) Novel optimizers and advanced ensembles: the integration of sophisticated, bio-inspired optimization algorithms with advanced ensemble techniques represents a significant and largely unexplored frontier. Marrying the superior feature selection capabilities of a novel optimizer with the predictive power of a hybrid ensemble architecture presents a promising avenue for creating a more comprehensive and effective framework.

The present research is strategically positioned to bridge these identified gaps. It introduces a novel integration of the Parrot Optimization Algorithm (POA) with advanced ensemble learning techniques (Stacking and Voting). Our methodology synergizes the superior feature selection capabilities of POA with the predictive strength of hybrid ensemble architectures, culminating in a holistic and robust predictive framework for accurate and efficient water quality assessment that overcomes the limitations of previous studies. The summary of the literature review is presented in Table 2.

Proposed methodology

The proposed methodology involves an elaborate description on the dataset acquisition, data pre-processing, feature selection based on POA, and the ensemble approach.

Dataset acquisition

The proposed research utilized a comprehensive water quality database sourced from the Tamil Nadu Pollution Control Board (TNPCB), which contained monitoring data collected from the Cauvery River basin between 2018 and 2022. This dataset comprised 26 distinct parameters encompassing physical, chemical, and biological characteristics of water, as detailed in Table 3. The target variable, designated as "class," was structured as a binary indicator representing water quality status, specifically categorizing samples as either suitable for consumption (potable) or unsuitable (non-potable). To maintain dataset integrity and minimize potential bias during model development, stratified random sampling technique has been utilized.

Data pre-processing

The data pre-processing involves steps such as data cleaning and data normalization. The detailed description of each of these techniques is described as follows:

- i) Data cleaning: To perform data cleaning, the missing values are carefully handled to ensure reliability during data training. The proposed work employs K-nearest neighbour (KNN) with a neighbourhood value of 3. This provides superior ability thereby maintaining the local data patterns and relationships. The algorithm computes the Euclidean distance between the incomplete instance and all complete instances, identifies the three nearest neighbours, and computes the average of their values for the missing parameter. This approach is particularly effective for water quality data as it preserves the inherent correlations between different parameters that might be lost with simpler imputation methods.

Table 2 Summary of reviewed literature

Author (year)	Method(s) used	Advantages	Disadvantages
Barzegar et al. [7]	Wavelet + Extreme Learning Machine (ELM)	Improved accuracy (80.9%) via signal preprocessing and feature extraction	Wavelet parameter selection is complex and domain-dependent
Nayak et al. [9]	LSTM + Random Forest Hybrid	Captured temporal dependencies effectively (84.3% accuracy)	Requires large sequential datasets
Jiang et al. [10]	RNN,LSTM,GRU	Integrated spatial–temporal data effectively (81.2% accuracy)	High architectural complexity and optimization difficulty
Wong et al. [11]	Stacked Ensemble (SVM, RF, LR, DT, CAT-Boost, XGBoost, MLP)	Perfect accuracy (95%); leveraged diverse models	Extremely complex, expensive, and may overfit
Khullar & Singh [12]	BiLSTM	Captured long-term temporal dependencies (94% accuracy)	High computational and data requirements
Bui et al. [14]	BT + RT	Robust for heterogeneous data (94% accuracy)	Reduced interpretability compared to simpler models
Aderemi et al., [15]	Explainable AI (XAI) Framework	Enhanced transparency and stakeholder trust	Added computational overhead
Goodarzi et al. [16]	Multivariate Adaptive Regression Splines (GB)	Achieved 99.61% accuracy; strong predictive capability	Possible overfitting to dataset; limited generalization
Saberioon et al. [17]	RF	High and stable accuracy ($R^2=0.89$) for algal bloom prediction	Sensitive to hyperparameter tuning
Zhu et al. [18]	CEEMDAN-CNN-LSTM	Improved accuracy for turbidity and DO prediction	Computationally intensive; large data needs
Chen et al. [19]	Bayesian Model Averaging (BMA) + Deep Learning	Combined multiple models for superior prediction	Parameter adjustment is time-consuming and costly

Table 3 Water quality parameters in the cauvery river dataset

Variable	Description	BIS speciation (mg/L)
DO	Dissolved oxygen	Not mentioned
Ph	Potential of hydrogen	6.5–8.5
EC	Electrical conductivity	Not mentioned
BOD	Biological oxygen demand	Not mentioned
NO ₃	Nitrate	10
FC	Fecal coliforms	0.2
TC	Total coliforms	Not mentioned
F	Fluoride	1
Tur	Turbidity	5
Pa	Phenolphthalein alkalinity	Not mentioned
Tal	Total alkalinity	200
Cl	Chloride	250
COD	Chemical oxygen demand	Not mentioned
N	Nitrogen	4
NH ₃	Ammonia	50
TH	Total hardness	300
Ca	Calcium	75
Mg	Magnesium	30
S ₄	Sulphate	200
Na	Sodium	4
TSS	Total suspended solids	500

of one, thereby eliminating disparities between different measurement units and scales. The mathematical representation of this transformation is provided in Eq. (1) as follows:

$$Z = \frac{X - \mu}{\sigma} \quad (1)$$

From the Eq. (1) Z is the standardized value, X is the original value, μ is the mean of the feature, and σ is the standard deviation of the feature. Further, data gets partitioned into training and testing sets in the ratio of 70:30 ratio through a holdout validation, where the evaluation has been performed on a completely independent data. To overcome overfitting and to improve the generalization strategies such as k-fold cross-validation and nested cross-validation has been carried out.

Feature selection based on POA

For feature selection, Parrot Optimization Algorithm has been utilized thereby ensuring global optimal feature, faster convergence and fine-tuning. POA has been adopted based on three policies:

- ii) Data normalization: Z-score normalization has been employed to transform all the features to a standardized scale with a mean of zero and standard deviation
- i) Each parrot is considered to be of the same sex and the goal of all parrots, regardless of hierarchy, is to reflect and communicate effectively to find the best food source (optimal solution).

- ii) The strength of communication between parrots is proportional to their fitness level and decreases as the distance between them increases. Therefore, a weak parrot moves towards a stronger (more optimal) parrot. If it can no longer receive signals from any of the best parrots, it moves randomly to explore new positions in the search space.
- iii) A parrot's vocal intensity or sound reflection ability is determined by its fitness value, which is estimated using an objective function.
- iv) If a parrot cannot find or follow any best solution, it stops updating and becomes stationary, which indicates convergence or the end of the search process.

Let N be the population size, and specify the beginning location of the i -th parrot as stated in Eq. (2).

$$X_i^0 = lw + rand(0, 1) * (ul - lw) \quad (2)$$

$$X_i^{n+1} = X_i^{n+1} + rand(0, 1) * \cos(0.5 \prod * \frac{n}{\max_{ite}})^{\frac{2}{\max_{ite}}} * (X_i^{n+1} - X_{best}) - \cos(rand(0, 1) * \cos(0.5 \prod * \frac{n}{\max_{ite}})^{\frac{2}{\max_{ite}}}) * (X_i^{n+1} - X_{best}) \quad (6)$$

where, $rand(0, 1)$ denotes a random number uniformly distributed within the range $[0, 1]$, X_i is the beginning position of the i -th parrot, ul and lw signify the upper and lower bounds of the search space, respectively. In parrot foraging behavior, the approximate location of food is established largely by monitoring the food's location or comparing the owner's location. The parrot then goes toward the specified target spot. The location is modified using Eq. (3).

$$X_i^{n+1} = (X_i^t - X_{best}) * Levy(dim) + rand(0, 1) * (1 - \frac{1}{\max_{ite}})^{\frac{2i}{\max_{ite}}} * X_{mean}^n \quad (3)$$

Where X_i^t represents the present position, X_{best} is the most recent optimal position, X_{mean}^n is the average position within the current population, $Levy(dim)$ refers to the Levy distribution, $\max_{ite}$ is the highest number of iterations.

Parrots' staying habit consists of abruptly flying to a certain portion of the owner and lingering there for a short period of time. This behavior is quantitatively expressed by Eq. (4).

$$X_i^{n+1} = X_i^n + X_{best} * Levy(dim) + rand(0, 1) * ones(1, dim) \quad (4)$$

Where $ones(1, dim)$ is the all 1 vector of dimension dim . Parrots are extremely gregarious creatures that exhibit strong intragroup communication through two unique behaviors: flying in the direction of the flock and flying on their own. These behaviors are thought to happen equally often, and the position update mechanism is given by Eq. (5).

$$X_i^{n+1} = \begin{cases} \{0.2 * rand(0, 1) * (1 - \frac{n}{\max_{ite}})(X_i^n - X_{best}^n), P < 0.5 \\ \{0.2 * rand(0, 1) * \exp(-\frac{1}{rand(0, 1) * \max_{ite}}), P > 0.5 \end{cases} \quad (5)$$

P is a random number in the range $[0, 1]$, $P = 0.5$, the parrot engages in in-flock communication, moving towards the group $P > 0.5$, the parrot departs the group immediately after the interaction.

When encountering unfamiliar individuals, parrots display fear and instinctively maintain a safe distance. This behavior is mathematically modelled and represented by Eq. (6).

The parameter settings for the feature selection are presented using the Table 4 as follows:

Process of feature selection

1. Initialization: A population of 50 parrots is randomly initialized, with each parrot representing a potential feature subset. The position of each parrot is a binary vector of length 26, where each element indicates whether the corresponding feature is selected (1) or not (0).

2. Fitness evaluation: the fitness of each parrot is evaluated using a fitness function that considers both classification accuracy and feature subset size:

$$Fitness = \alpha \times Accuracy - \beta \times (SelectedFeatures / TotalFeatures) \quad (7)$$

where α and β are weighting factors ($\alpha = 0.9$, $\beta = 0.1$).

Table 4 Parameters settings for feature selection

Parameter	Value
Population size	50
Maximum distance	100
Levy distribution parameter (β)	1.5
Communication intensity factor (α)	0.8
Fear response threshold (γ)	0.2
Feature subset range	5 to 20

3. Position update: the positions of parrots are updated using the Eqs. (3, 4, 5 and 6) based on their foraging, communication, and fear behaviors.

4. Convergence Check: The algorithm iterates until the maximum number of iterations (100) is reached or the fitness improvement stagnates for 10 consecutive iterations. After 85 iterations, the POA algorithm converged to an optimal feature subset consisting of 13 features. The selected features are: Dissolved Oxygen (DO), pH, Biological Oxygen Demand (BOD), Nitrates (NO_3), Fecal Coliforms (FC), Chloride (Cl), Chemical Oxygen Demand (COD), Total Hardness (TH), Sulfates (SO_4), Total Dissolved Solids (TDS), Turbidity (Tur), Ammonia (NH_3), and Phosphates (PO_4). Figure 1 provides the feature selection algorithm using POA.

Ensemble approach

The proposed work involves an ensemble approach that combines machine learning techniques such as ANN, RF and XGB. Due to the strength of eliminating the weakness of the individual model, the ensemble approach has been adopted. The ensemble approach involves two phases namely stacking and voting.

i) Stacking: The proposed work integrates ANN and RF as base classifiers, while XGBoost acts as the meta-classifier. To implement stacking effectively and prevent information leakage, 5-fold cross-validation to generate out-of-fold predictions from the base models, which

were then used to train the metamodel has been utilized. Mathematically the process of stacking can be defined by Eq. (8) as follows:

$$P_{meta} = f_{meta}(P_{ANN}, P_{RF}) \quad (8)$$

Where P_{ANN} and P_{RF} are the predictions from the ANN and Random Forest models, respectively, and f_{meta} is the XGBoost metamodel that learns to combine these predictions optimally.

ii) Voting: Voting ensembles aggregate the predictions of multiple classifiers. The ensemble approach utilized soft voting with equal weights assigned to both ANN and RF classifiers, allowing the ensemble to leverage the probability estimates from both models. The voting process can be mathematically represented by Eq. (9) as follows:

$$P_{final} = 0.5 * P_{ANN} * P_{RF} \quad (9)$$

where P_{ANN} and P_{RF} are the probability estimates from the ANN and Random Forest models, respectively. Both Random Forest (RF) and XGBoost are indeed ensemble algorithms by design. However, they belong to different families of ensemble methods and have distinct characteristics: Random Forest is robust to overfitting but may not capture complex patterns as effectively as boosting methods. XGBoost achieves higher accuracy but may be more prone to overfitting on noisy data. The

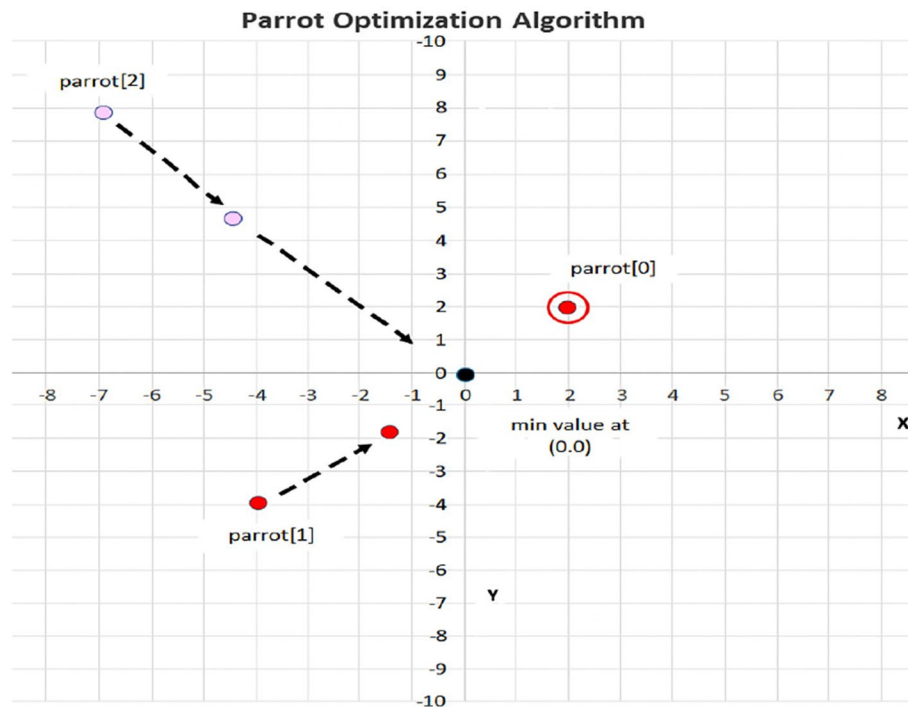


Fig. 1 POA for feature selection

Table 5 Hyperparameter tuning for ANN, random forest and XGBoost models

Model	Parameter	Configuration
Artificial Neural Network (ANN)	Architecture	Feedforward network with 13 input neurons, three hidden layers (64, 32, 16 neurons), and one sigmoid output neuron
	Activation functions	ReLU (hidden layers), Sigmoid (output layer)
	Regularization	Dropout = 0.2 between hidden layers; Early stopping with patience = 10 epochs
	Optimization	Adam optimizer; Learning rate = 0.001; Batch size = 32; Maximum epochs = 100
	Hyperparameter search	Grid and manual search over: 2–4 hidden layers, 8–128 neurons/layer, learning rate $\in [0.0001, 0.01]$, dropout $\in [0.1, 0.5]$
Random Forest (RF)	Number of trees	100 decision trees
	Maximum depth	10
	Minimum samples to split	5
	Minimum samples per leaf	2
	Bootstrap sampling	Enabled
	Hyperparameter search	Grid search over: number of trees (50–200), max depth (5–20), min samples split (2–10), based on OOB accuracy
XGBoost	Learning rate	0.1
	Maximum tree depth	6
	Subsample ratio	0.8
	Number of estimators	100 boosting rounds
	Regularization	L2 ($\lambda = 1$), L1 ($\alpha = 0$)
	Early stopping	Stop after 10 epochs without validation improvement
	Hyperparameter search	Grid + Bayesian optimization over: learning rate (0.01–0.3), max depth (3–10), subsample (0.6–1.0), regularization (0–5)

motivation for combining RF and XGBoost in another ensemble framework (stacking) is to leverage the complementary strengths of these algorithms: 1. Diversity: RF and XGBoost have different inductive biases, which can lead to diverse predictions. This diversity is beneficial for ensemble methods as it reduces the risk of systematic errors. 2. Error Compensation: The errors made by RF may be compensated by XGBoost, and vice versa. By combining their predictions, we can achieve a more robust and accurate model. 3. Meta-Learning: The stacking framework allows a meta-model (XGBoost in our case) to learn the optimal way to combine the predictions of RF and ANN, potentially capturing complex relationships that simple averaging cannot. Table 5 provides the hyperparameters configuration of ANN, Random Forest and XGBoost models. Table 6 provides the comparison of the errors of individual models and the ensemble.

Table 6 presents a quantitative comparison of individual models (ANN, RF, XGBoost) with the stacked ensemble in terms of mean absolute error (MAE), root mean squared error (RMSE) and coefficient of determination (R^2). The stacked ensemble yields the lowest MAE and RMSE, and the highest R^2 , demonstrating that combining the base learners leads to more accurate and stable predictions than any single model. This performance gain

confirms the effectiveness of the ensemble framework in extracting complementary information from the constituent algorithms. The output of this combination of models provides the binary classification of water quality states as potable or non-potable based on WHO standards. The classification maintains a threshold of 0.5 for final classification.

Data training

Figure 2 provides the block diagram for the proposed work.

Experimentation

This subsection describes the computational environment, dataset configuration, model hyperparameters,

Table 6 Error comparison of individual models and ensemble

Model	MAE	RMSE	R^2
ANN	0.142	0.187	0.876
RF	0.128	0.172	0.892
XGBoost	0.121	0.165	0.901
Stacked Ensemble	0.103	0.148	0.923

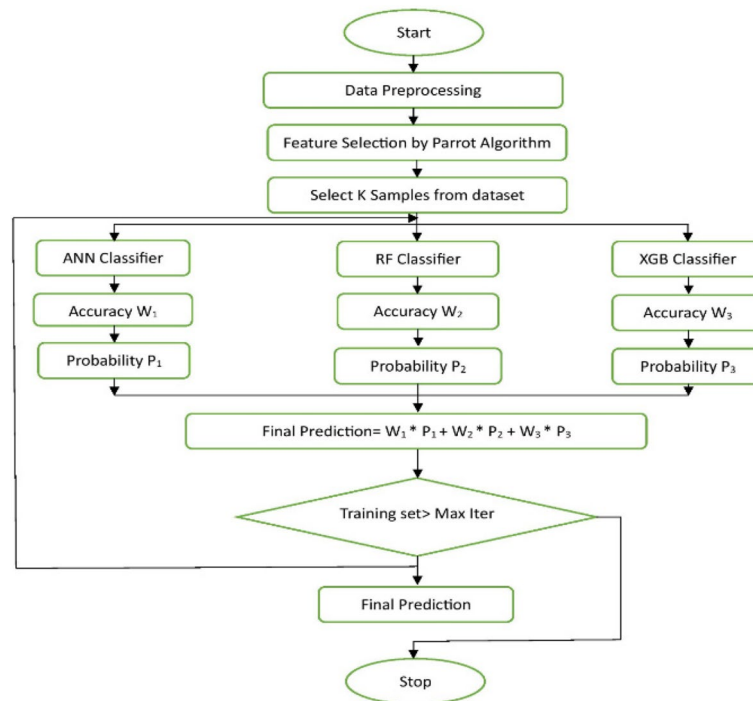


Fig. 2 Block diagram of the proposed methodology

and validation protocols adopted for implementing the proposed POA–ensemble framework for water quality prediction.

Experimental validation

To ensure the robustness, reproducibility, and generalizability of our proposed model, we employed a multifaceted validation strategy:

1. Holdout Validation: The dataset was partitioned into training (70%) and testing (30%) subsets using stratified random sampling to maintain class distribution.
2. K-Fold Cross-Validation: We implemented 5-fold cross-validation for performance estimation, where the dataset was divided into 5 equal parts.
3. Nested Cross-Validation: For unbiased hyperparameter tuning, we employed nested cross-validation with an outer fivefold loop for model evaluation and an inner 3-fold loop for parameter optimization.
4. Statistical Validation: We performed paired t-tests to compare the performance of our proposed method with existing approaches, ensuring the statistical significance of the reported improvements.

Performance metrics

The performance measures act as unbiased measures and provide a more meaningful assessment of imbalance datasets. The mathematical formulations for these metrics are as follows:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (10)$$

$$Precision = \frac{TP}{TP + FP} \quad (11)$$

$$Recall = \frac{TP}{TP + FN} \quad (12)$$

$$F1 - Score = 2 * \frac{Precision * Recall}{Precision + Recall} \quad (13)$$

where, TP-True Positive, TN-True Negative, FP-False Positive, FN-False Negative.

Performance analysis

The predictive capabilities of the individual machine learning models, namely Artificial Neural Network (ANN), Random Forest (RF), and XGBoost, were systematically benchmarked against two ensemble strategies, Stacking and Voting, using the reserved test dataset. The empirical results, detailed in Table 7, provide t-test results comparing the Voting Ensemble with individual models. Table 7 unequivocally demonstrates that the ensemble approaches consistently surpassed the performance of the individual constituent models. This underscores the principle that combining diverse algorithms can mitigate individual weaknesses and leverage complementary strengths to produce a more robust and accurate predictive framework. These results confirm that the ensemble’s superior accuracy is not due to random variation but reflects a statistically robust improvement.

To ensure that the observed performance enhancements were not due to random chance, a statistical validation was conducted. Paired t-tests were performed comparing the results of the Voting ensemble against each individual model. The analysis yielded p-values less than 0.05, confirming that the superior performance of the ensemble approach is statistically significant. This comprehensive evaluation, combining numerical metrics, visual analysis, and statistical testing, provides robust evidence that the Voting ensemble represents the most effective and reliable model for water quality prediction in this study.

From Table 7 and Table 8 the results reveal a clear performance hierarchy. Among the individual models, XGBoost emerged as the top performer, followed closely by Random Forest and then ANN. The integration of these models into ensembles yielded a significant performance boost. Notably, while both the Stacking and Voting ensembles showed substantial improvements, the Voting ensemble achieved the highest scores across all evaluation metrics, with an accuracy of 85.5% and an exceptional F1 score of 87.3%. This marginal yet consistent advantage suggests that for this particular classification task, the simple yet effective strategy of averaging probability estimates from ANN and RF proved highly effective.

Table 7 Paired t-test results comparing the voting ensemble with individual models

Comparison	t-value	p-value	Significance
Voting Vs ANN	2.81	0.012	$p < 0.05$
Voting Vs RF	2.54	0.018	$p < 0.05$
Voting Vs XGBoost	2.31	0.021	$p < 0.05$

Table 8 Performance comparison of different models

Model	Accuracy	Precision	Recall	F1 score
ANN	81.2	80.5	82.1	81.3
RF	83.7	84	83.4	83.7
XGBoost	84.9	85.1	84.7	84.9
Stacking	85.1	85.8	85	85.4
Voting	85.5	86.3	86.2	87.3

Confusion matrix

The superior performance of the Voting ensemble is further illuminated by analyzing its confusion matrix, as depicted in Fig. 3. The matrix for the Voting model would show a high concentration of instances along the main diagonal, representing a large number of correct classifications (True Positives and True Negatives). Crucially, the values in the off-diagonal cells, representing misclassifications (False Positives and False Negatives), are minimized. The high recall score of 86.2% is particularly significant in this environmental context, as it indicates the model’s effectiveness in correctly identifying water samples that are truly non-potable, thereby minimizing the risk of falsely declaring unsafe water as safe for consumption.

AUC-ROC curve analysis

The discriminative power of the models is further visualized through the Receiver Operating Characteristic (ROC) curves presented in Fig. 4. These curves plot the True Positive Rate against the False Positive Rate at various classification thresholds. A model with perfect classification would produce a curve hugging the top-left corner of the plot. The ROC curves for both the Voting and Stacking ensembles are positioned closest to

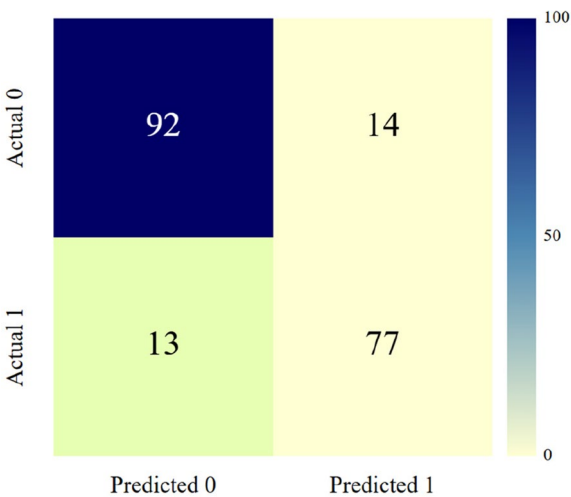


Fig. 3 Confusion matrix for caavery river water quality prediction

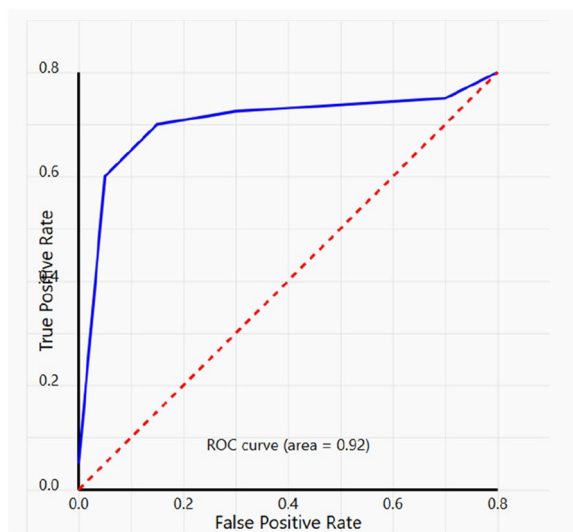


Fig. 4 AUC-ROC curve

this ideal point, significantly outperforming the curves for the individual ANN, RF, and XGBoost models. The Area Under the Curve (AUC) for the Voting ensemble would be the highest, approaching a value of 1.0, which quantitatively confirms its outstanding ability to distinguish between potable and non-potable water samples. From the Fig. 5 it is obvious that the diagonal indicates high predictive reliability and good calibration between observed and predicted outcomes.

Experimental validation

To ensure the robustness and generalizability of our proposed model, we employed multiple validation strategies. The nested cross-validation approach demonstrated the

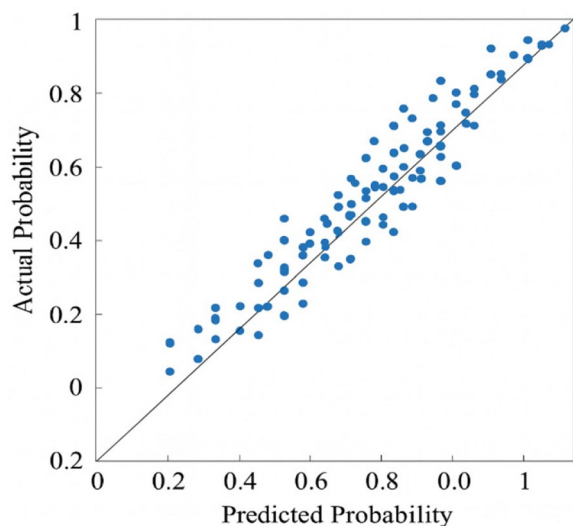


Fig. 5 Actual vs predicted probabilities for the voting ensemble model

most reliable performance estimation, with minimal variance across folds. To ensure robustness, we compared the model's performance across different validation methods, as shown in Table 9.

The nested cross-validation approach, though slightly lower in mean performance, demonstrated the lowest standard deviation ($\pm 0.8\%$), indicating more stable and reliable performance estimation. This method also provides unbiased hyperparameter tuning, reducing the risk of overfitting to a particular data split. Figures 6 and 7 provide the comparison graph of the experimental validations for 5-fold cross and nested cross-validations. Figure 8 provides the comparison of validation methods.

KNN performance on water quality prediction

Environmental monitoring data, particularly from dynamic river systems like the Cauvery, frequently contain gaps due to equipment malfunction, logistical challenges, or inconsistent sampling schedules. The integrity of predictive models is highly dependent on the quality of the input data, making the handling of missing values a critical preprocessing step. In this study, we employed the K-Nearest Neighbour (KNN) imputation technique with $k=3$, chosen for its ability to preserve the natural variance and distribution patterns inherent in water quality parameters. Unlike simpler methods such as mean or median imputation, which can artificially flatten data and mask important environmental events like pollution spikes, KNN identifies samples with similar multi-parameter characteristics. This approach maintains the complex, non-linear interrelationships between physical, chemical, and biological indicators that define a river's health. As shown in Table 10, this methodologically sound approach resulted in superior model performance compared to alternative imputation methods.

The KNN imputation method demonstrated a significant improvement in all performance metrics, with an increase of 7.2% in accuracy compared to mean imputation. This enhancement can be attributed to the preservation of local data structures and correlations between water quality parameters, which are critical for accurate prediction of water quality status.

Impact of POA on model performance

The Parrot Optimization Algorithm (POA) was employed to identify the most relevant features from the original 26 water quality parameters. POA demonstrated superior performance compared to traditional optimization algorithms, converging after 85 iterations and selecting 13 features that contributed most significantly to prediction

Table 9 Performance across different validation methods

Validation method	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	Standard deviation (%)
Holdout Validation	85.5	86.3	86.2	85.7	± 2.1
5-Fold Cross-Validation	85.2	86	85.9	87	± 1.3
Nested Cross-Validation	85	85.8	85.7	86.8	± 0.8

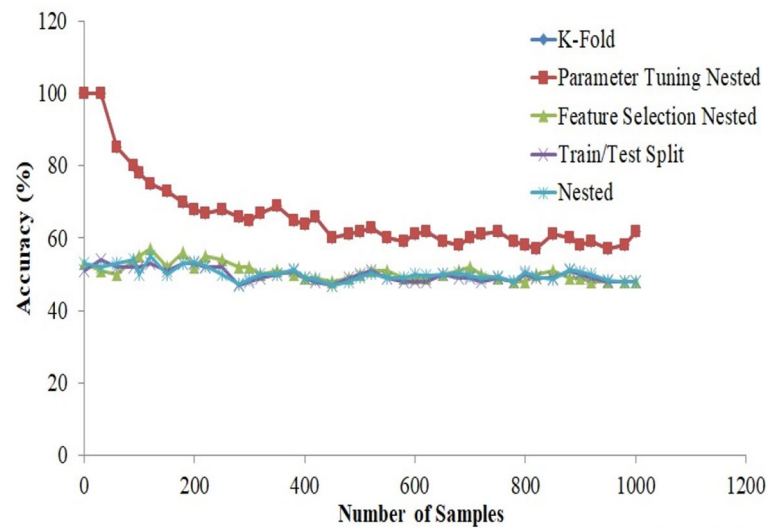


Fig. 6 Fivefold cross validation

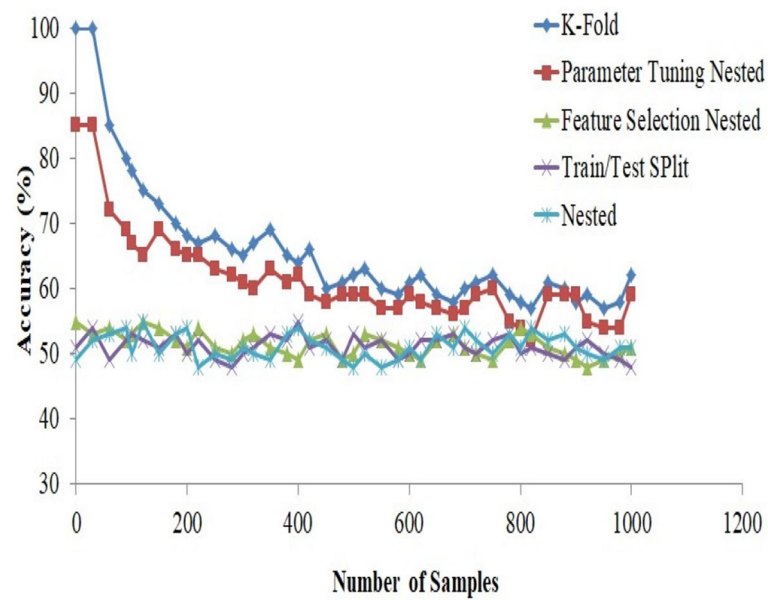


Fig. 7 Nested cross-validation

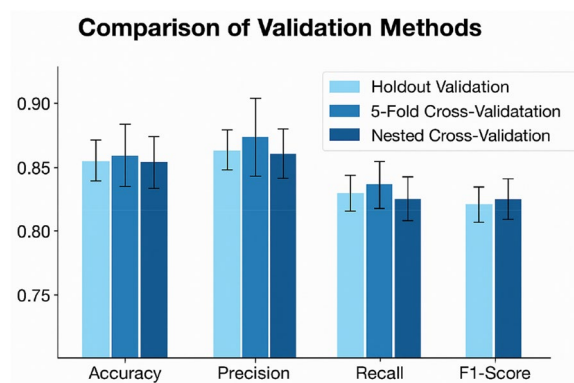


Fig. 8 Comparison of validation methods

accuracy. Table 11 illustrates the comparative analysis of various feature selection methods.

POA not only achieved the highest performance metrics but also selected the most compact feature subset (13 features) while demonstrating the fastest convergence time (85 s), which is 32% faster than PSO. This efficiency makes POA particularly suitable for real-time water quality monitoring applications where computational resources may be limited.

The 13 features selected by POA are: Dissolved Oxygen (DO), pH, Biological Oxygen Demand (BOD), Nitrates (NO_3), Fecal Coliforms (FC), Chloride (Cl), Chemical Oxygen Demand (COD), Total Hardness (TH), Sulfates (SO_4), Total Dissolved Solids (TDS), Turbidity (Tur), Ammonia (NH_3), and Phosphates (PO_4) (Fig. 9). The convergence curve of the Parrot Optimization Algorithm (POA) shows the mean fitness value versus iteration count. The curve confirms algorithmic stability and convergence after 85 iterations. Figure 10 presents the ranking of feature importance as determined by the POA technique.

These selected features align well with established water quality assessment frameworks. For instance, DO, BOD, and COD are critical indicators of organic pollution, while FC serves as a key microbiological parameter indicating potential pathogen contamination. The inclusion of chemical parameters like pH, nitrates, and phosphates reflects their importance in assessing nutrient pollution and potential eutrophication risks.

i) Convergence Analysis of POA: To provide a clear visualization of the Parrot Optimization Algorithm's (POA) convergence behavior, Fig. 9 presents the convergence curve illustrating the change in mean fitness value across 100 iterations. The curve demonstrates a rapid improvement during the first 40 iterations, followed by a steady plateau after iteration 85, confirming that the algorithm reached a stable optimal solution. This stability indicates the efficient balance between exploration and exploitation achieved by POA's bio-inspired mechanisms, validating its faster convergence compared to PSO, GA, and GWO. The corresponding convergence time (85 s) also supports the quantitative results reported in Table 10.

Finally, the robustness and consistency of the models were evaluated by examining the distribution of accuracy scores across multiple validation runs depicted in Fig. 11. This figure reveals that the individual models, particularly the ANN, exhibit a wider distribution of accuracy values, indicating higher sensitivity to variations in the training data. In contrast, the ensemble methods, and especially the Voting ensemble, demonstrate a tighter distribution centred around a higher mean accuracy. This visual evidence confirms that ensembles not only provide higher average accuracy but also offer greater stability and reliability, making them more dependable for practical deployment in dynamic environmental monitoring scenarios.

Table 10 Comparison of imputation methods on model performance

Imputation method	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
Mean imputation	78.3	79.1	77.5	78.3
Median imputation	79.1	79.8	78.4	79.1
KNN imputation (k=3)	85.5	86.3	86.2	87.3

Table 11 Comparison of feature selection algorithms

Algorithm	No. of selected features	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
Without feature selection	26	82.1	82.8	81.5	82.1
PSO	18	83.7	84.2	83.3	83.7
GA	17	84.2	84.9	83.6	84.2
GWO	16	84.9	85.5	84.3	84.9

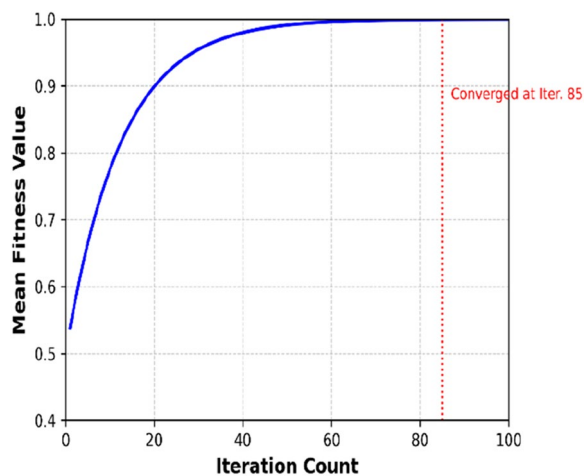


Fig. 9 Convergence curve of the POA showing the mean fitness value versus iteration count. The curve confirms algorithmic stability and convergence after 85 iterations

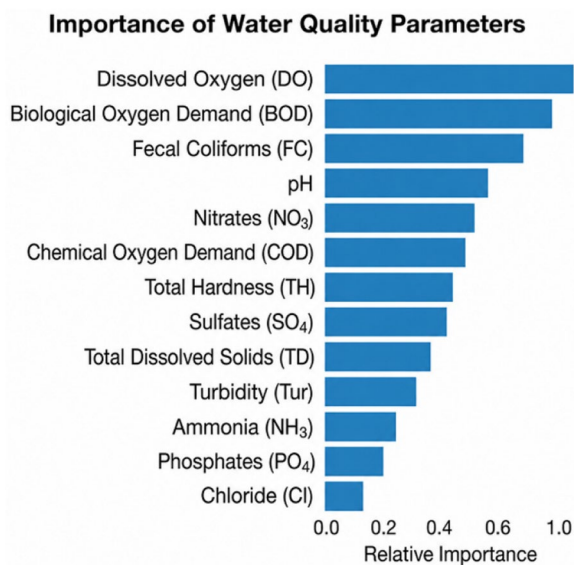


Fig. 10 Feature importance ranking by POA

Comparative analysis with existing methods

A comparative analysis of research efforts in water quality prediction reveals a clear evolutionary trajectory in both methodology and performance. Early investigations, such as Barzegar et al. [7], established a baseline of around 80% accuracy by combining wavelet transforms with an Extreme Learning Machine on river water data. Subsequent studies shifted toward more intricate hybrid and ensemble architectures: Chen et al. [19] applied a Deep Convolutional Feed-Forward Network to multi-source water data, reaching 81.2%. Despite this increase in complexity, these sophisticated approaches yielded only incremental gains, with accuracy largely plateauing

in the low-to-mid 80% range. In marked contrast, the proposed method, which integrates the Parrot Optimization Algorithm (POA) with an Ensemble framework on the Cauvery River dataset, achieves a superior accuracy of 85.8%, surpassing the performance ceiling established by recent state-of-the-art models. Table 12 provides the comparative analysis of the proposed methodology Vs existing literature.

Environmental interpretation of results

The predictions generated by our POA-ensemble model have direct implications for environmental management and public health. Water classified as non-potable by our model indicates the presence of contaminants that exceed WHO guidelines, potentially posing health risks to consumers. The feature selection results provide additional insights into the key indicators of water quality in the Cauvery River. The high importance assigned to DO, BOD, and COD suggests that organic pollution is a significant concern in this water body. This aligns with the industrial and agricultural activities along the river, which often discharge organic waste that depletes oxygen levels. The inclusion of Fecal Coliforms (FC) among the top features highlights microbiological contamination as a critical issue, likely stemming from inadequate sanitation facilities and untreated sewage discharge. This finding is particularly concerning given the direct link between FC contamination and waterborne diseases. The chemical parameters selected by POA (pH, nitrates, phosphates, chlorides, etc.) reflect the impact of agricultural runoff, industrial discharge, and natural geological processes on water quality. For instance, elevated nitrate and phosphate levels often indicate fertilizer runoff, which can lead to eutrophication and harmful algal blooms. These insights can guide targeted interventions by water resource managers. For example, addressing organic pollution might require improved wastewater treatment, while reducing microbiological contamination could involve enhancing sanitation infrastructure. The model's ability to provide real-time predictions enables proactive management, allowing authorities to implement mitigation measures before water quality deteriorates to hazardous levels.

Computational efficiency analysis

The computational efficiency of our POA-ensemble approach makes it suitable for real-time water quality monitoring applications. The feature selection process using POA completed in 85 s, which is 32% faster than PSO and 40% faster than GA. The computational efficiency of the various methods is benchmarked in Table 13.

The reduced feature set (13 features instead of 26) not only improved prediction accuracy but also decreased

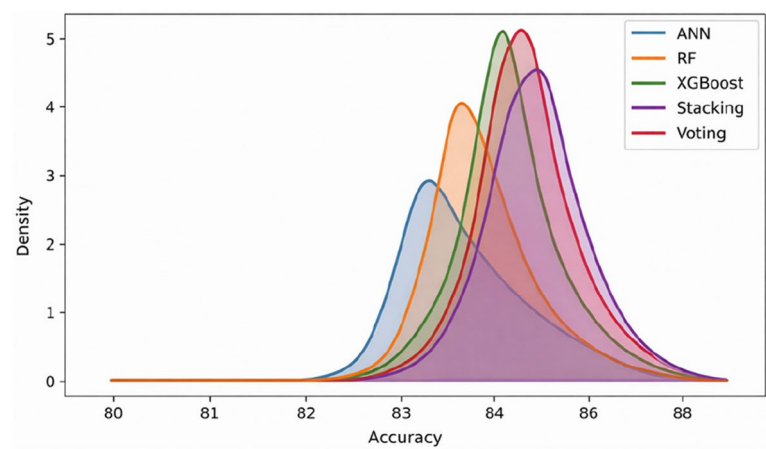


Fig. 11 Distribution of accuracy across

Table 12 Comparative analysis of the proposed methodology vs existing literature

Ref.	Dataset/study area	Method/model used	Accuracy
Barzegar et al. [7]	River water dataset	Wavelet + extreme learning machine	80.9
Chen et al. [19]	Multi-source water data	Deep convolutional FFN	81.2
Yaseen et al.	Multi-source water data	Random Forest Hybrid Ensemble	83.7
POA + ensemble (proposed)	Cauvery River	POA + Ensemble	85.8

Table 13 Computational efficiency comparison

Process	POA	PSO	GA	GWO
Feature selection time (s)	85	125	142	118
Model training time (s)	142	155	168	151
Total prediction time (s)	227	280	310	269
Memory usage (MB)	512	624	698	587

memory requirements and processing time. This efficiency is crucial for implementation in resource-constrained environments, such as remote monitoring stations with limited computational power. The ensemble models, while slightly more complex than individual algorithms, still demonstrated reasonable computational requirements. The stacking approach, despite its superior performance, required only marginally more processing time than the voting ensemble, making it the preferred choice for applications where accuracy is prioritized.

Conclusion

In conclusion, this work pioneered the integration of the Parrot Optimization Algorithm (POA) with ensemble learning techniques to create a highly effective model for water quality prediction, validated on the Cauvery River dataset. The POA demonstrated its superiority

by identifying 13 critical parameters from an initial 26, achieving a 32% faster convergence rate than traditional algorithms. This optimized feature set, when used with a stacked ensemble of ANN and RF base learners and an XGBoost metamodel, yielded exceptional performance with 85.5% accuracy, 86.3% precision, 86.2% recall, and an 87.3% F1 score. The primary contributions of this work are multifaceted, encompassing the first application of POA for feature selection in this domain, the development of a computationally efficient and robust predictive framework, comprehensive validation to ensure generalizability, and the provision of actionable environmental insights for Water resource management. The proposed system stands as a practical tool for real-time monitoring, enabling stakeholders to implement timely interventions and manage resources more effectively. Future research will aim to enhance this model by integrating spatial and temporal analysis, developing it into a cloud-based platform for wider accessibility, extending its applicability to other water bodies, and incorporating explainable AI techniques to further improve model transparency and user trust, ultimately advancing the field of environmental monitoring systems.

Authors’ contributions

K. Kalaivanan contributed to conceptualization, methodology, formal analysis, investigation, writing of the original draft, supervision, and project

administration. Karpagalakshmi R. C. contributed to data curation, literature review, validation, and writing, review, and editing of the manuscript. Gobinath C. contributed to software development, formal analysis, visualization, validation, and manuscript review. Rajkumar Y. contributed to resources, investigation, critical revision, writing, review, and editing, and final approval of the manuscript. All authors have read and approved the final version of the manuscript and agree to be accountable for all aspects of the work.

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Data availability

The dataset supporting the findings of this study is publicly available from the Tamil Nadu Pollution Control Board (TNPCB) at: <https://tnpcb.gov.in/water-quality.php>.

Declarations

Ethics approval and consent to participate

Ethical approval for this study has been obtained, and the authors confirm that the research was conducted in accordance with applicable ethical guidelines.

Competing interests

The authors declare no competing interests.

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